

TurnRD Per-Turn Credit Assignment — v3 R9 Policy

Greedy $K=1$ rollouts on AlfWorld held-out tasks. Each turn shows the observation on the first line and the agent’s action on the second; only the action is shaded by per-turn credit (green = positive, red = negative) because the action is what receives reward / blame. Color saturation is normalized per trajectory. The most-decisive turn in successes (largest positive credit) and the most-fatal turn in failures (largest negative credit) are marked with ★. Turn 0 is the initial observation (uncolored, no credit). By construction, per-turn credits sum to the trajectory’s final reward R via the v -projection.

Successes

Task 6504 — Success ($R = +1.00$)

Showing Turn 0 + turns 0–11 of 14 (window around the decisive turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a bed 1, a drawer 10, a drawer 9, a drawer 8, a drawer 7, a drawer 6, a drawer 5, a drawer 4, a drawer 3, a drawer 2, a drawer 1, a dresser 1, a garbagecan 1, a shelf 9, a shelf 8, a shelf 7, a shelf 6, a shelf 5, a shelf 4, a shelf 3, a shelf 2, and a shelf 1. Your task is to: look at statue under the desk lamp. (*initial*)

Action: look

Turn 1: You are in the middle of a room. Looking quickly around you, you see nothing.

Action: go to shelf 1 (*credit* = -0.026)

Turn 2: You arrive at shelf 1. On the shelf 1, you see a alarmclock 1, and a statue 1.

Action: take statue 1 from shelf 1 (*credit* = +0.873) ★ (*decisive*)

Turn 3: You pick up the statue 1 from the shelf 1.

Action: go to shelf 2 (*credit* = -0.247)

Turn 4: You arrive at shelf 2. On the shelf 2, you see a pen 1.

Action: go to shelf 3 (*credit* = -0.137)

Turn 5: You arrive at shelf 3. On the shelf 3, you see nothing.

Action: go to shelf 4 (*credit* = -0.120)

Turn 6: You arrive at shelf 4. On the shelf 4, you see a houseplant 1.

Action: go to shelf 5 (*credit* = -0.034)

Turn 7: You arrive at shelf 5. On the shelf 5, you see a pencil 2.

Action: go to shelf 6 (*credit* = +0.037)

Turn 8: You arrive at shelf 6. On the shelf 6, you see nothing.

Action: go to shelf 7 (*credit* = +0.017)

Turn 9: You arrive at shelf 7. On the shelf 7, you see a pencil 3.

Action: go to shelf 8 (*credit* = -0.036)

Turn 10: You arrive at shelf 8. On the shelf 8, you see nothing.

Action: go to shelf 9 (*credit* = -0.051)

Turn 11: You arrive at shelf 9. On the shelf 9, you see nothing.

Action: go to garbagecan 1 (*credit* = +0.009)

⋮ (*turns 12–13 elided*)

Task 6500 — Success ($R = +1.00$)

Showing Turn 0 + turns 23–34 of 37 (window around the decisive turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a bed 1, a desk 1, a drawer 5, a drawer 4, a drawer 3, a drawer 2, a drawer 1, a dresser 1, a garbagecan 1, a shelf 12, a shelf 11, a shelf 10, a shelf 9, a shelf 8, a shelf 7, a shelf 6, a shelf 5, a shelf 4, a shelf 3, a shelf 2, a shelf

1, and a sidetable 1. Your task is to: look at book under the desk lamp. (*initial*)

Action: look

⋮ (*turns 1–22 elided*)

Turn 23: You are facing the drawer 2, drawer 1, and dresser 1. Next to it, you see nothing.

Action: go to shelf 1 (*credit* = -0.168)

Turn 24: You arrive at shelf 1. On the shelf 1, you see a vase 1.

Action: go to drawer 2 (*credit* = -0.665)

Turn 25: You arrive at drawer 2. On the drawer 2, you see nothing.

Action: go to drawer 3 (*credit* = -0.355)

Turn 26: You arrive at drawer 3. The drawer 3 is closed.

Action: open drawer 3 (*credit* = +1.031)

Turn 27: You open the drawer 3. The drawer 3 is open. In it, you see a pencil 3.

Action: close drawer 3 (*credit* = -1.020)

Turn 28: You close the drawer 3.

Action: go to drawer 4 (*credit* = -0.267)

Turn 29: You arrive at drawer 4. The drawer 4 is closed.

Action: open drawer 4 (*credit* = +1.152) ★ (*decisive*)

Turn 30: You open the drawer 4. The drawer 4 is open. In it, you see nothing.

Action: close drawer 4 (*credit* = -0.840)

Turn 31: You close the drawer 4.

Action: go to drawer 5 (*credit* = +0.330)

Turn 32: You arrive at drawer 5. The drawer 5 is closed.

Action: open drawer 5 (*credit* = +1.152)

Turn 33: You open the drawer 5. The drawer 5 is open. In it, you see nothing.

Action: close drawer 5 (*credit* = -0.698)

Turn 34: You close the drawer 5.

Action: go to desk 1 (*credit* = +0.123)

⋮ (*turns 35–36 elided*)

Task 6507 — Success (R = +1.00)

Showing Turn 0 + turns 0–11 of 19 (window around the decisive turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a bed 1, a drawer 10, a drawer 9, a drawer 8, a drawer 7, a drawer 6, a drawer 5, a drawer 4, a drawer 3, a drawer 2, a drawer 1, a dresser 1, a garbagecan 1, a shelf 7, a shelf 6, a shelf 5, a shelf 4, a shelf 3, a shelf 2, and a shelf 1. Your task is to: examine the pillow with the desk lamp. (*initial*)

Action: look

Turn 1: You are in the middle of a room. Looking quickly around you, you see nothing.

Action: go to bed 1 (*credit* = +0.142)

Turn 2: You arrive at bed 1. On the bed 1, you see a book 1, a laptop 2, a laptop 1, a pillow 3, a pillow 2, and a pillow 1.

Action: take pillow 3 from bed 1 (*credit* = +1.055) ★ (*decisive*)

Turn 3: You pick up the pillow 3 from the bed 1.

Action: go to shelf 1 (*credit* = +0.002)

Turn 4: You arrive at shelf 1. On the shelf 1, you see nothing.

Action: move pillow 3 to shelf 1 (*credit* = -1.069)

Turn 5: Nothing happens.

Action: go to shelf 2 (*credit* = +0.077)

Turn 6: You arrive at shelf 2. On the shelf 2, you see a pen 1.
Action: go to shelf 3 (*credit = +0.155*)

Turn 7: You arrive at shelf 3. On the shelf 3, you see nothing.
Action: go to shelf 4 (*credit = +0.138*)

Turn 8: You arrive at shelf 4. On the shelf 4, you see a creditcard 3.
Action: go to shelf 5 (*credit = +0.147*)

Turn 9: You arrive at shelf 5. On the shelf 5, you see a alarmclock 1, a creditcard 2, a window 2, and a window 1.
Action: look (*credit = +0.325*)

Turn 10: You are facing the shelf 7, shelf 6, and shelf 5. Next to it, you see nothing.
Action: go to shelf 2 (*credit = -0.179*)

Turn 11: You arrive at shelf 2. On the shelf 2, you see a pen 1.
Action: go to shelf 6 (*credit = -0.200*)

⋮ (*turns 12-18 elided*)

Failures

Task 6524 — Failure (R = +0.00)

Showing Turn 0 + turns 10-21 of 40 (window around the fatal turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a bed 1, a desk 1, a drawer 17, a drawer 16, a drawer 15, a drawer 14, a drawer 13, a drawer 12, a drawer 11, a drawer 10, a drawer 9, a drawer 8, a drawer 7, a drawer 6, a drawer 5, a drawer 4, a drawer 3, a drawer 2, a drawer 1, a dresser 1, a garbagecan 1, a shelf 6, a shelf 5, a shelf 4, a shelf 3, a shelf 2, and a shelf 1. Your task is to: look at bowl under the desklamp. (*initial*)

Action: look
 ⋮ (*turns 1-9 elided*)

Turn 10: You arrive at dresser 1. On the dresser 1, you see a alarmclock 1, a cd 1, a creditcard 2, a desklamp 1, and a keychain 2.

Action: use desklamp 1 (*credit = +1.059*)

Turn 11: You turn on the desklamp 1.

Action: take alarmclock 1 from dresser 1 (*credit = +0.668*)

Turn 12: You pick up the alarmclock 1 from the dresser 1.

Action: use desklamp 1 (*credit = +0.481*)

Turn 13: You turn on the desklamp 1.

Action: move alarmclock 1 to dresser 1 (*credit = +0.484*)

Turn 14: You move the alarmclock 1 to the dresser 1.

Action: go to drawer 1 (*credit = -0.346*)

Turn 15: You arrive at drawer 1. The drawer 1 is closed.

Action: open drawer 1 (*credit = +0.916*)

Turn 16: You open the drawer 1. The drawer 1 is open. In it, you see nothing.

Action: close drawer 1 (*credit = -1.012*) ★ (*fatal*)

Turn 17: You close the drawer 1.

Action: go to drawer 2 (*credit = +0.130*)

Turn 18: You arrive at drawer 2. The drawer 2 is closed.

Action: open drawer 2 (*credit = +0.980*)

Turn 19: You open the drawer 2. The drawer 2 is open. In it, you see nothing.

Action: close drawer 2 (*credit = -0.908*)

Turn 20: You close the drawer 2.

Action: go to drawer 3 (*credit* = +0.225)

Turn 21: You arrive at drawer 3. On the drawer 3, you see nothing.

Action: go to drawer 4 (*credit* = -0.165)

⋮ (*turns 22–39 elided; trajectory ran to step limit without reaching the goal*)

Task 6514 — Failure (R = +0.00)

Showing Turn 0 + turns 22–33 of 40 (window around the fatal turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a armchair 1, a cabinet 1, a drawer 21, a drawer 20, a drawer 19, a drawer 18, a drawer 17, a drawer 16, a drawer 15, a drawer 14, a drawer 13, a drawer 12, a drawer 11, a drawer 10, a drawer 9, a drawer 8, a drawer 7, a drawer 6, a drawer 5, a drawer 4, a drawer 3, a drawer 2, a drawer 1, a dresser 1, a garbagecan 1, a sidetable 5, a sidetable 4, a sidetable 3, a sidetable 2, a sidetable 1, and a sofa 1. Your task is to: look at box under the desk lamp. (*initial*)

Action: look

⋮ (*turns 1–21 elided*)

Turn 22: You close the drawer 2.

Action: go to drawer 3 (*credit* = +0.019)

Turn 23: You arrive at drawer 3. The drawer 3 is closed.

Action: open drawer 3 (*credit* = +0.987)

Turn 24: You open the drawer 3. The drawer 3 is open. In it, you see nothing.

Action: close drawer 3 (*credit* = -0.907)

Turn 25: You close the drawer 3.

Action: go to drawer 4 (*credit* = -0.078)

Turn 26: You arrive at drawer 4. On the drawer 4, you see nothing.

Action: go to drawer 5 (*credit* = -0.324)

Turn 27: You arrive at drawer 5. The drawer 5 is closed.

Action: open drawer 5 (*credit* = +0.761)

Turn 28: You open the drawer 5. The drawer 5 is open. In it, you see nothing.

Action: close drawer 5 (*credit* = -1.066) ★ (*fatal*)

Turn 29: You close the drawer 5.

Action: go to drawer 6 (*credit* = -0.168)

Turn 30: You arrive at drawer 6. The drawer 6 is closed.

Action: open drawer 6 (*credit* = +0.822)

Turn 31: You open the drawer 6. The drawer 6 is open. In it, you see nothing.

Action: close drawer 6 (*credit* = -0.977)

Turn 32: You close the drawer 6.

Action: go to drawer 7 (*credit* = +0.093)

Turn 33: You arrive at drawer 7. The drawer 7 is closed.

Action: open drawer 7 (*credit* = +1.064)

⋮ (*turns 34–39 elided; trajectory ran to step limit without reaching the goal*)

Task 6541 — Failure ($R = +0.00$)

Showing Turn 0 + turns 28–39 of 40 (window around the fatal turn).

Turn 0: -= Welcome to TextWorld, ALFRED! -= You are in the middle of a room. Looking quickly around you, you see a bed 1, a desk 1, a drawer 2, a drawer 1, a garbagecan 1, a shelf 1, and a sidetable 1. Your task is to: examine the alarmclock with the desklamp. (*initial*)

Action: look

⋮ (*turns 1–27 elided*)

Turn 28: You turn on the desklamp 1.

Action: move laptop 1 to desk 1 (*credit = -0.282*)

Turn 29: You move the laptop 1 to the desk 1.

Action: take book 1 from desk 1 (*credit = -0.469*)

Turn 30: You pick up the book 1 from the desk 1.

Action: use desklamp 1 (*credit = -0.739*)

Turn 31: You turn on the desklamp 1.

Action: move book 1 to desk 1 (*credit = +0.529*)

Turn 32: You move the book 1 to the desk 1.

Action: go to bed 1 (*credit = -0.487*)

Turn 33: You arrive at bed 1. On the bed 1, you see a book 2, a cellphone 2, a cellphone 1, a pillow 2, a pillow 1, and a teddybear 1.

Action: take book 2 from bed 1 (*credit = +0.952*)

Turn 34: You pick up the book 2 from the bed 1.

Action: go to desk 1 (*credit = -0.771*)

Turn 35: You arrive at desk 1. On the desk 1, you see a book 1, a desklamp 1, a laptop 1, a mug 1, a pen 2, a pen 1, a pencil 3, a pencil 2, and a tissuebox 1.

Action: use desklamp 1 (*credit = -0.538*)

Turn 36: You turn on the desklamp 1.

Action: move book 2 to desk 1 (*credit = +1.052*)

Turn 37: You move the book 2 to the desk 1.

Action: take laptop 1 from desk 1 (*credit = +0.311*)

Turn 38: You pick up the laptop 1 from the desk 1.

Action: use desklamp 1 (*credit = -0.279*)

Turn 39: You turn on the desklamp 1.

Action: use desklamp 1 (*credit = -0.803*) ★ (*fatal*)